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Mathematical Foundations of Data Science (Probability)

Lecture 03

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Random Variables

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

Definition (Random Variables)

A function $X : \Omega \rightarrow \mathbb{R}$ is called a **random variable** with respect to $\mathcal{F}$ if:

$$\{\omega \in \Omega : X(\omega) \leq x\} \in \mathcal{F} \quad \forall x \in \mathbb{R}.$$

Remarks:

- A random variable is neither random nor a variable; it is a deterministic function
- The definition of a random variable is closely tied to the underlying σ -algebra $\mathcal{F}$
A function X that is a random variable with respect to $\mathcal{F}$ may not be a random variable with respect to $\mathcal{G}$ for $\mathcal{G} \neq \mathcal{F}$
- If X is a random variable with respect to $\mathcal{F}$, it is called an **$\mathcal{F}$ -measurable function**
- The definition of a random variable does not involve the underlying probability measure $\mathbb{P}$

Examples

- $\Omega = \{1, 2, \dots, 6\}$, $\mathcal{F} = \{\emptyset, \Omega\}$, $X(\omega) = \omega$
Is X a random variable with respect to $\mathcal{F}$?

Examples

- $\Omega = \{1, 2, \dots, 6\}, \quad \mathcal{F} = \left\{ \emptyset, \Omega \right\}, \quad X(\omega) = \omega$

Is X a random variable with respect to $\mathcal{F}$?

- What functions X are random variables with respect to $\mathcal{F}$?

Examples

- $\Omega = \{1, 2, 3, 4, 5\}$, $\mathcal{F} = \left\{ \emptyset, \Omega, A, A^c \right\}$ for a fixed $A \subseteq \Omega$

What functions X are random variables with respect to $\mathcal{F}$?

Examples

- $\Omega = \{1, 2, 3, 4, 5\}, \quad \mathcal{F} = \sigma \left(\left\{ \{1\}, \{2, 3\} \right\} \right)$

What functions X are random variables with respect to $\mathcal{F}$?

Examples

- $\Omega = \mathbb{N}$, $\mathcal{F} = 2^\Omega$

What functions X are random variables with respect to $\mathcal{F}$?

Whenever $\mathcal{F} = 2^\Omega$, **every function** $X : \Omega \rightarrow \mathbb{R}$ **is a random variable** with respect to $\mathcal{F}$.

Cumulative Distribution Function (CDF)

Cumulative Distribution Function (CDF)

Definition (Cumulative Distribution Function)

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

Given a random variable $X : \Omega \rightarrow \mathbb{R}$ with respect to $\mathcal{F}$, its **cumulative distribution function (CDF)**

$F_X : \mathbb{R} \rightarrow [0, 1]$ is defined as

$$F_X(x) = \mathbb{P}(\{\omega \in \Omega : X(\omega) \leq x\}), \quad x \in \mathbb{R}.$$

Remarks on notation:

- $\{\omega \in \Omega : X(\omega) \leq x\} = \{X \leq x\}$
- $\mathbb{P}(\{\omega \in \Omega : X(\omega) \leq x\}) = \mathbb{P}(\{X \leq x\})$

Properties of CDF

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$

Theorem 1 (Properties of CDF)

Let $X : \Omega \rightarrow \mathbb{R}$ be a random variable with respect to $\mathcal{F}$ with CDF F_X .
Then, F_X satisfies the following properties.

- Monotonicity:** If $x \leq y$, then $F_X(x) \leq F_X(y)$.
- $\lim_{x \rightarrow -\infty} F_X(x) = 0, \quad \lim_{x \rightarrow +\infty} F_X(x) = 1.$
- Right-Continuity:** F_X is right-continuous at every point in its domain, i.e., for all $x \in \mathbb{R}$,

$$\lim_{\varepsilon \downarrow 0} F_X(x + \varepsilon) = F_X(x).$$

- For any $x \in \mathbb{R}$, $\lim_{\varepsilon \downarrow 0} F_X(x - \varepsilon) = \mathbb{P}(\{X < x\})$.
- F_X is continuous at a point $x \in \mathbb{R}$ if and only if $\mathbb{P}(\{X = x\}) = 0$

Proof of Theorem 1: Part 1

- Choose arbitrary x, y such that $x \leq y$
- Note that

$$\mathbb{F}_X(x) = \mathbb{P}(\{X \leq x\}), \quad \mathbb{F}_X(y) = \mathbb{P}(\{X \leq y\}).$$

- Because $x \leq y$, it follows that

$$\{X \leq x\} \subseteq \{X \leq y\}.$$

- Using monotonicity property of $\mathbb{P}$, it follows that

$$\mathbb{P}(\{X \leq x\}) \leq \mathbb{P}(\{X \leq y\}).$$

Proof of Theorem 1: Part 2

- Suppose $x_1, x_2, \dots$ is **monotone decreasing** sequence such that

$$x_1 \geq x_2 \geq \dots, \quad \lim_{n \rightarrow \infty} x_n = -\infty.$$

- Then, we have

$$\{X \leq x_1\} \supseteq \{X \leq x_2\} \supseteq \dots$$

- Using continuity of probability measure $\mathbb{P}$,

$$\lim_{x \rightarrow -\infty} F_X(x) = \lim_{n \rightarrow \infty} F_X(x_n) = \lim_{n \rightarrow \infty} \mathbb{P}(\{X \leq x_n\}) = \mathbb{P}\left(\bigcap_{n \in \mathbb{N}} \{X \leq x_n\}\right) = \mathbb{P}(\emptyset) = 0.$$

- Exercise:** Along similar lines as above, show that

$$\lim_{x \rightarrow +\infty} F_X(x) = 1.$$

Proof of Theorem 1: Part 3

- Choose an arbitrary $x \in \mathbb{R}$
- Suppose that $x_1, x_2, \dots$ is any **monotone decreasing sequence** such that

$$x_1 \geq x_2 \geq \dots \geq x, \quad \lim_{n \rightarrow \infty} x_n = x.$$

- Then, note that

$$\{X \leq x_1\} \supseteq \{X \leq x_2\} \supseteq \dots \supseteq \{X \leq x\}, \quad \bigcap_{n \in \mathbb{N}} \{X \leq x_n\} = \{X \leq x\}.$$

- Using the continuity of probability measure $\mathbb{P}$, we then have

$$\begin{aligned} \lim_{\varepsilon \downarrow 0} F_X(x + \varepsilon) &= \lim_{n \rightarrow \infty} F_X(x_n) = \lim_{n \rightarrow \infty} \mathbb{P}(\{X \leq x_n\}) = \mathbb{P}\left(\bigcap_{n \in \mathbb{N}} \{X \leq x_n\}\right) \\ &= \mathbb{P}(\{X \leq x\}) = F_X(x). \end{aligned}$$

Proof of Theorem 1: Part 4

- Choose an arbitrary $x \in \mathbb{R}$

Suppose that $x_1, x_2, \dots$ is any **monotone increasing sequence** such that

$$x_1 \leq x_2 \leq \dots \leq x, \quad \lim_{n \rightarrow \infty} x_n = x.$$

- Then, note that

$$\{X \leq x_1\} \subseteq \{X \leq x_2\} \subseteq \dots, \quad \bigcup_{n \in \mathbb{N}} \{X \leq x_n\} = \{X < x\}.$$

- Using the continuity of the probability measure $\mathbb{P}$,

$$\begin{aligned} \lim_{\varepsilon \downarrow 0} F_X(x - \varepsilon) &= \lim_{n \rightarrow \infty} F_X(x_n) = \lim_{n \rightarrow \infty} \mathbb{P}(\{X \leq x_n\}) = \mathbb{P}\left(\bigcup_{n \in \mathbb{N}} \{X \leq x_n\}\right) \\ &= \mathbb{P}(\{X < x\}). \end{aligned}$$

Proof of Theorem 1: Part 5

- Choose an arbitrary $x \in \mathbb{R}$, and observe that

$$\{X \leq x\} = \{X < x\} \sqcup \{X = x\}.$$

- Therefore, we get

$$\mathbb{P}(\{X \leq x\}) = \mathbb{P}(\{X < x\}) + \mathbb{P}(\{X = x\}).$$

- Equivalently, we have

$$F_X(x) = \mathbb{P}(\{X < x\}) + \mathbb{P}(\{X = x\}).$$

- For F_X to be continuous at the point x , it must satisfy both left-continuity and right-continuity at the point x , i.e.,

$$\lim_{\varepsilon \downarrow 0} F_X(x + \varepsilon) = \lim_{\varepsilon \downarrow 0} F_X(x - \varepsilon) \iff F_X(x) = \mathbb{P}(\{X < x\}).$$

Example

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

- Let X be a random variable with CDF F_X as follows:

$$F_X(x) = \begin{cases} 0, & x < 0, \\ 0.5, & 0 \leq x < 1, \\ \frac{x}{2}, & 1 \leq x < 2, \\ 1, & x \geq 2. \end{cases}$$

Sketch the above CDF, and compute the following.

- $\mathbb{P}(\{X \leq 0.8\})$.
- $\mathbb{P}(\{X \geq 0.25\})$.
- $\mathbb{P}(\{X = 0\})$.
- $\mathbb{P}(\{X \in (1, 3)\})$

CDF of a random variable X provides a complete probabilistic description of X .

Discrete and Continuous Random Variables

Discrete Random Variable

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

Definition (Discrete Random Variable)

A random variable $X : \Omega \rightarrow \mathbb{R}$ is said to be **discrete** if there exists a **finite or countably infinite** set $\mathcal{X} \subset \mathbb{R}$ such that

$$\sum_{x \in \mathcal{X}} \mathbb{P}(\{X = x\}) = 1.$$

Remark:

- For any “nice” set $B \subseteq \mathbb{R}$,

$$\mathbb{P}(\{X \in B\}) = \sum_{x \in B} \mathbb{P}(\{X = x\}).$$

Here, “nice” means that $\{X \in B\} \in \mathcal{F}$ so that we can $\mathbb{P}(\{X \in B\})$ makes sense
In measure theory, such nice sets are called **Borel sets**

Probability Mass Function (PMF)

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

Definition (Probability Mass Function)

Let $X : \Omega \rightarrow \mathbb{R}$ be a random variable.

The **probability mass function (PMF)** of X is the function $p_X : \mathbb{R} \rightarrow [0, 1]$ defined as

$$p_X(x) = \mathbb{P}(\{X = x\}), \quad x \in \mathbb{R}.$$

Remark:

- If X is a discrete random variable taking values in a discrete set $\mathcal{X}$, then

$$\sum_{x \in \mathcal{X}} p_X(x) = 1.$$

- If X is a discrete random variable taking values in a discrete set $\mathcal{X}$, then its CDF and PMF satisfy

$$F_X(x) = \mathbb{P}(\{X \leq x\}) = \sum_{\substack{y \in \mathcal{X}: \\ y \leq x}} p_X(y), \quad x \in \mathbb{R}.$$

- For a discrete random variable X , its PMF p_X provides the complete probabilistic description

Examples of PMFs

- Bernoulli(p), $p \in [0, 1]$

$$\mathcal{X} = \{0, 1\},$$

$$p_X(x) = \begin{cases} p, & x = 1, \\ 1 - p, & x = 0, \\ 0, & \text{otherwise.} \end{cases}$$

- Uniform($\{1, \dots, n\}$), $n \in \mathbb{N}$

$$\mathcal{X} = \{1, \dots, n\},$$

$$p_X(x) = \begin{cases} \frac{1}{n}, & x \in \{1, \dots, n\}, \\ 0, & \text{otherwise.} \end{cases}$$

Examples of PMFs

- Geometric(p), $p \in [0, 1]$

$$\mathcal{X} = \mathbb{N}, \quad p_X(x) = \begin{cases} p(1-p)^{x-1}, & x \in \mathbb{N}, \\ 0, & \text{otherwise.} \end{cases}$$

- Binomial(n, p), $n \in \mathbb{N} \cup \{0\}$, $p \in [0, 1]$

$$\mathcal{X} = \{0, \dots, n\}, \quad p_X(x) = \begin{cases} \binom{n}{x} p^x (1-p)^{n-x}, & x \in \{0, \dots, n\}, \\ 0, & \text{otherwise.} \end{cases}$$

Examples of PMFs

- $\text{Poisson}(\lambda)$, $\lambda > 0$

$$\mathcal{X} = \{0, 1, 2, \dots\},$$

$$p_X(x) = \begin{cases} e^{-\lambda} \frac{\lambda^x}{x!}, & x \in \{0, 1, 2, \dots\}, \\ 0, & \text{otherwise.} \end{cases}$$

- A custom PMF:

$$\mathcal{X} = \{1, 2, \dots\},$$

$$p_X(x) = \begin{cases} \frac{6}{\pi^2} \frac{1}{x^2}, & x \in \{1, 2, \dots\}, \\ 0, & \text{otherwise.} \end{cases}$$